



Upstream  
Digital Center  
Leading Digital Excellence

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saudi aramco



## Appendix A to SIB

# Saudi Aramco Seismic Processing Compute Expansion Project

Technical Specifications

UPSTREAM DIGITAL CENTER

Release V1.13



## GENERAL INSTRUCTIONS

1. Bidder shall provide an introduction of their company and list of key partners participating in providing the proposed solution, e.g. data center manufacturer, supercomputer manufacturer and integrator(s).
2. Bidder shall note that this project will be conducted in two phases:
  - **Phase-1:** supply and commission of the managed HPC Infrastructure as a Service with supercomputing capabilities that meet or exceed the technical specifications in this document and provides minimum FP64 performance below:
    - Theoretical High Performance Linpack (HPL) performance ( $R_{peak}$ ) of 145PFlops
    - Sustained HPL performance ( $R_{max}$ ) of 80PFlops of GPU nodes
  - **Phase-2:** *provisional option* to be executed, approximately one year after the deployment of Phase-1 to expand Phase-1 supercomputing capabilities with additional resources that meet or exceed Phase-2 technical specifications in this document and provides minimum FP64 High Performance Linpack (HPL) performance below:
    - Theoretical HPL performance ( $R_{peak}$ ) of 55PFlops
    - Sustained HPL performance ( $R_{max}$ ) of 35PFlops of GPU nodes
3. Bidder shall confirm that the execution of Phase-2 of this project is at Saudi Aramco's sole discretion. Provisional requirements such as this option for Phase-2 may or may not be ordered in response to business requirements.
4. Bidder shall confirm via a letter from the OEM in a format acceptable to Saudi Aramco, that it has the support of the OEM for the duration of the Contract and any extensions thereof.

# Seismic Processing Compute Expansion Project

Phase I

145 Petaflops

## REMOTE COMPUTING FACILITY REQUIREMENTS

### General Requirements

1. The term **Remote Computing Facility** include, without limitation: land, data center space, chillers, fencing, securing perimeter, CCTV cameras, power and cooling as well as office spaces to be used by the Bidder-related support requirements.
2. The **Remote Compute Facility** shall comply with the Uptime Institute's **Tier-1 standards** (at minimum). The Bidder shall supply certification of this compliance as per the terms and conditions of the Contract.
3. All Bidders are required to provide documented proof of either ownership or lease agreement of the proposed **Remote Computing Facility**. This documentation should demonstrate the Bidder's legal right to operate and provide services from the specified location. Bidder shall include a copy of the property deed, lease agreement, or any other relevant legal document verifying the Bidder's rights to the **Remote Computing Facility** as part of the proposal. Failure to provide this proof may result in the rejection of the proposal.
4. Bidder should acknowledge that the new **Remote Computing Facility** shall operate independently of the existing DAMMAM-7 facility and not be affected in the event of DAMMAM-7 facility decommissioning or power-off.
5. Bidders are encouraged to provide energy efficient data centers to align with the Kingdom's vision of promoting sustainability and reducing environmental impact. Bidders shall provide estimated PUE (Power Usage Effectiveness) metric for their proposed solution assuming IT equipment is working at full capacity.
6. Bidder shall provide a project timeline for resource mobilization, system deployment, and configuration from the award date. The Bidder must designate a project manager within one week of the award date. The project manager will be responsible for developing a project plan document and a logically linked, resource-loaded, level 2 schedule with Saudi Aramco in accordance with the submitted timeline. Additionally, the project manager shall provide Saudi Aramco with a weekly progress update.

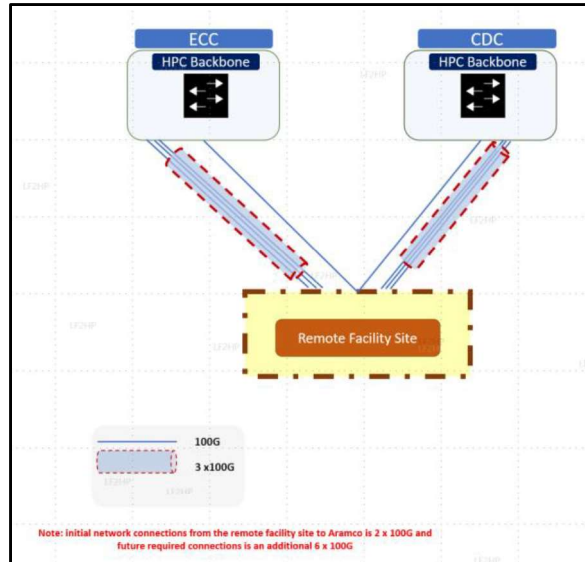
### Physical Security

7. Bidder shall be fully responsible for the physical security of the facility, the equipment inside it, and the surrounding perimeter on 24X7 basis. This shall include equipment supplied by the Bidder, Saudi Aramco, and any third party.
8. Bidder shall supply, implement, maintain, and operate a physical security setup to manage accessibility to the **Remote Computing Facility**, monitor and record all personnel activities within the **Remote Computing Facility** and the outside perimeter.
9. Bidder shall be able to issue required permits to allow physical accessibility to the **Remote Computing Facility** for all required personnel all the times (24X7).
10. All physical security related data including but not limited to authorization data, monitoring records, and logs shall be accessible to Saudi Aramco representatives real-time/online, kept for a period of a minimum of one (1) year, and shall be available for Saudi Aramco for review and reference as needed. Bidder shall explain the mechanism.
11. Bidder shall supply, install, and commission to production a physical security access management solution including but not limited to surveillance cameras, all doors. The Bidder shall ensure the high availability and business continuity for the provided solution.

12. Bidder shall install enough monitoring cameras to monitor the **Remote Computing Facility's** surrounding perimeters, all doors and entrances of the **Remote Computing Facility**, data rooms, aisles, storage and common areas. The cameras shall be installed in a way to allow facial recognition and identification of all the personnel attending the monitored areas. In addition, Bidder shall install enough cameras to monitor the movement of all personnel within the data rooms, aisles, storage and common rooms. The cameras shall be installed in a way to allow tracking the movement of the personnel between different areas.
13. Bidder shall also implement a secure door system (Electronic keycards/access badges) on all the **Remote Computing Facility's** main entrances including but not limited to, all entrances to the data rooms, and all entrances to the network equipment rooms. All these entrances shall also be equipped with cameras and sensors to detect tailgating. In case of a tailgating incident, these entrances shall alarm, send an alert message through physical security system, and take a snapshot of the tailgater clear enough to identify his/her face.
14. Bidder shall ensure that access logs and authorization data to be recorded on 24X7 basis.
15. Bidder shall man guard the **Remote Computing Facility** on 24X7 basis and shall build a scheduled security tour inside the **Remote Computing Facility** and around the outside perimeter.
16. Access to the site shall be restricted to authorized personnel only. All personnel shall be properly identified with a current valid ID allowing access to restricted areas based on need.
17. Physical protection against damage from fire, flood, earthquake, explosion, civil unrest, and other forms of natural or man-made disaster shall be designed and applied for critical computer and communications facilities and resources.

### Network Connection to Upstream Digital Center (UDC) Network

18. Saudi Aramco shall provide the fiber cable between Saudi Aramco UDC data center and the connection point (Meet-me Room) at the Bidder proposed **Remote Computing Facility**. However, the Bidder is fully responsible to/shall provide all required networking equipment at both sites that is needed to provide IP network connectivity between UDC and the provided solution (Compute and Storage).
19. Bidder shall describe how they plan to integrate the proposed HPC system with Saudi Aramco existing network, with a specific focus on the data movement between **Remote Computing Facility** and the existing DAMMAM-7 and Saudi Aramco data centers managed by UDC.
20. Bidder shall be responsible for the management, configuration, and security protection of network devices end point at their end throughout the contract period.
21. Bidder shall be responsible to provide all required expertise to ensure reliable connection from their side throughout the contract period.
22. Bidder shall note that lease payment will not be processed till the network connection is established between their proposed solution and UDC network.
23. Bidder shall note below high level network diagram which shows UDC-to-**Remote Computing Facility** connectivity:



24. Bidder shall confirm the supply of the below networking hardware/software requirements at Saudi Aramco facilities,

| SKU                       | Description                                                                                        | Quantity |
|---------------------------|----------------------------------------------------------------------------------------------------|----------|
| <b>PTX10K-LC1202-36MR</b> | PTX10K 4.8Tbps Line Card - 32 port QSFP28 + 4 port QSFP-DD (JNP10K-LC1202)                         | 2        |
| <b>S-PTX10K-400G-P1-P</b> | SW, PTX10K, 400G, Pre1, without SW support, Perpetual                                              | 12       |
| <b>SVC-ND-PTX-L1202</b>   | Juniper Care Next Day Support for PTX10K-LC1202-36MR                                               | 2        |
| <b>SVC-COR-PTX-400GP1</b> | Juniper Care Core Support for S-PTX10K-400G-P1-P                                                   | 12       |
| <b>QSFP-100G-LR4-C</b>    | QSFP28, 100GBASE-LR4, SMF 10 km, Standard Temperature (0 through 70 DEGREE C), Duplex LC connector | 8        |

## INTEGRATION SERVICES

25. Bidder shall provide a plan for the integration of the proposed HPC systems with existing Saudi Aramco infrastructure.
26. Bidder shall describe the integration plan of the proposed HPC system with existing network, particularly the data movement between **Remote Computing Facility** and the existing DAMMAM-7 and Saudi Aramco data centers.
27. Bidder shall provide details on the data transfer protocols and encryption methods employed to ensure secure data movement.
28. Bidder shall confirm and guarantee that provided resources (Compute, network, and storage) are dedicated for Saudi Aramco, and no one else has access to the resources even if not utilized.
29. Bidder shall either develop or provide a **Data Copy Tool** that integrates with Saudi Aramco Seismic Projects Data in the existing DAMMAM-7 and Saudi Aramco data centers. The Bidder



within the first 90 days from contract award date shall work closely with the Company Representatives to collect application requirements, complete detailed design of the application, and submitting detailed project scope of work with target dates to be approved by Company Representative. The Bidder shall complete this application within 60 days from approval date by the Company Representatives.

30. Bidder shall develop a Graphical User Interface application to manage Seismic Projects in the proposed **Remote Computing Facility**, named **Seismic Projects Management Application**, and integrates with Saudi Aramco Seismic Projects databases. The Bidder, within the first 90 days from contract award date, shall work closely with the Company Representative to collect application requirements, complete detailed design of the application, and submit detailed project scope of work with target dates to be approved by the Company Representatives. The Bidder shall complete this application within 60 days from approval date by the Company Representative.
31. Bidder shall develop or provide a Graphical User Interface application for managing HPC Seismic Jobs in the proposed **Remote Computing Facility** that integrates with the supercomputer Job Scheduler. The Bidder, within the first 90 days from contract award date, shall work closely with the Company Representative to collect application requirements, complete detailed design of the application, and submitting detailed project scope of work with target dates to be approved by Company Representative. The Bidder shall complete this application within 60 days from approval date by the Company Representative.
32. Bidder can provide 29-31 as one application, as approved by Company Representative.

## SUPERCOMPUTER TECHNICAL REQUIREMENTS

### General Technical Requirements

33. All hardware supplied, including, but not limited to, CPU and GPU nodes, **should not have any power or system limitations/capping.** The supplied hardware shall be capable of being fully utilized to the maximum rated power capacity, to ensure potential future application optimization can be realized.
34. Bidder shall provide an integrated supercomputing system that provides (at minimum) FP64 HPL performance below:
  - Theoretical HPL (High Performance Linpack) performance ( $R_{peak}$ ) of **145 PFlops**, calculated as aggregate sum of  $R_{peak}$  of CPU nodes and  $R_{peak}$  of GPU nodes,
  - Sustained HPL performance ( $R_{max}$ ) of **80PFlops** of GPU nodes.
35. Bidder shall acknowledge that providing a solution that does not meet the above performance requirements and acceptance tests will require providing additional hardware till all requirements are met or waived by Saudi Aramco.
36. The proposed hardware shall be **certified and approved** by application owners (SLB, and AspenTech) and/or Saudi Aramco.
37. All required software licenses to meet the supercomputing requirements shall be included in the proposal for the full duration of the contract without any additional cost to Saudi Aramco after the start of the lease agreement.
38. All memory channels on the provided **Compute (CPU and/or GPU)**, Windows, Visualization and General-Purpose nodes shall be fully populated, with minimum DIMM size of 32GB DDR5 unless approved by Saudi Aramco.
39. No limit on the ingress and egress network traffic between the **Remote Computing Facility** and Saudi Aramco, and existing Dammam-7 facilities. Saudi Aramco shall have no cap on the data volume moved between the proposed **Remote Computing Facility** and its own facilities.



## Compute Technical Requirements

### GPU COMPUTE

40. Bidder shall provide at least **560 GPU** nodes, each node with below **minimum** specifications:

- **GPU:**
  1. 4xNVIDIA H100 SXM GPUs with at least 80GB HBM3 GPU memory delivering at least 3.35TB/s GPU memory bandwidth
- **CPU:** below options can be supplied
  1. 2x AMD Genoa CPUs, each with 64 physical cores with total DDR5 RAM capacity that is at least 2X total HBM capacity per node (**768GB or higher**), or
  2. 2x Intel Sapphire Rapids CPUs, each with 52 physical cores with total DDR5 RAM capacity that is at least 2X total HBM capacity per node (**768GB or higher**)
- **Local boot disk:** 1x480GB (or higher) SAS SSD RI use, 1DWPD
- **Local data disk:** usable capacity of 14TB NVMe (or higher) (RAID-0 on multiple drives, e.g. Micron 7450) providing minimum read of 16GB/s per local filesystem
- **Network:** 400Gbps (100Gbps / GPU) (or more if needed to meet acceptance tests)
- **Operating System:** Red Hat Enterprise Linux (RHEL)

41. Based on key HPC applications performance benchmarks done on 4-way and 8-way nodes, it was found that an 8-way GPU node is 1.4 the performance of 4-way GPU nodes. Therefore, [\(as alternative to above\)](#) Bidder may provide at least **400 8-way GPU** nodes with below minimum specifications:

- **GPU:**
  - 8xNVIDIA H100 SXM GPUs with at least 80GB HBM3 GPU memory delivering at least 3.35TB/s GPU memory bandwidth
- **CPU:** below options can be supplied
  - 2x AMD Genoa CPUs, each with 64 physical cores with total DDR5 RAM capacity that is at least 2X total HBM capacity per node (**1280GB or higher**), or
  - 2x Intel Sapphire Rapids CPUs, each with 52 physical cores with total DDR5 RAM capacity that is at least 2X total HBM capacity per node (**1280GB or higher**)
- **Local boot disk:** 1x480GB (or higher) SAS SSD RI use, 1DWPD
- **Local data disk:** below options can be supplied
  - usable capacity of 24TB NVMe (or higher) (RAID-0 on multiple drives, e.g. Micron 7450) providing minimum read of 24GB/s per local filesystem
  - Bidder can provide a **central storage** solution that is reliable and secure and can provide dedicated 24TB per GPU node that delivers minimum read of 3GB/s per GPU. The central storage shall be based on NVMe drives with ability to handle aggregate performance of simultaneous I/O from all GPU nodes and maintain the per GPU capacity and performance requirement.
- **Network:** 800Gbps (100Gbps / GPU) (or more if needed to meet acceptance tests)
- **Operating System:** Red Hat Enterprise Linux (RHEL)

### CPU-ONLY LINUX COMPUTE

42. Bidder shall provide at least **560 CPU** nodes, each node with below minimum specifications:

- **CPU:** 2xAMD Genoa CPUs, each with 96 physical cores, with 1536GB RAM
- **Local boot disk:**
  1. 1x480GB (or higher) SAS SSD RI use, 1DWPD.
  2. Diskless boot service can be accepted given that the Bidder explains the limitation of not having local boot disk, assurance that no difference in system performance, provide references of previous use cases in production of similar setup, and provide details of expected boot time for all CPU nodes simultaneously using proposed solution
- **Local data disk:** usable capacity of 14TB NVMe (or higher) (RAID-0 on multiple drives, e.g. Micron 7450) providing minimum read of 10GB/s per local filesystem
- **Network:** 200Gbps (or higher)
- **Operating System:** Red Hat Enterprise Linux (RHEL)

#### VISUALIZATION NODES

43. The visualization solution is targeted to be used by maximum **96 concurrent users**.

44. Bidder shall provide at least sixteen (16) nodes, each with below minimum specifications (or equivalent):

- **CPU:** 2xAMD Genoa CPUs, each with 64 physical cores, with 1536GB RAM
- **GPU:** 2xNVIDIA L40 GPUs PCIe5 (include licenses to be used by multiple users concurrently)
- Full stack remote visualization software (FastX, NICE, or equivalent)
- The nodes can be configured as Linux or Windows nodes running on virtual environment. Bidder shall provide a virtualization hypervisor to segment servers and manage the configured VMs as per Saudi Aramco requirements.
- **Local boot disk:** 1x480GB (or higher) SAS SSD RI use, 1DWPD
- **Local data disk:** 2x6TB (or higher) NVMe (RAID-0 on multiple drives)
- **Network:** 100Gbps (or higher)
- **Parallel Storage access:** required for both windows and linux operating systems
- **Operating System:** Red Hat Enterprise Linux (RHEL) and Microsoft Windows Server supported by SLB Petrel Application.

45. Bidder may [propose alternative](#) solutions as long as they meet the below requirements:

- Support simultaneous provisioning of **96 concurrent users** accessing windows or Linux VMs supported by SLB Petrel Application.
- Full stack remote visualization software (PC over IP, FastX, NICE, or equivalent)
- 256 GB DDR5 RAM per user
- 20 X86 physical cores per user
- Enough local disk for the hypervisor and VMs OS
- L40 or L4 GPUs that can be sliced to support multiple concurrent sessions
- 12TB local NVMe per every 8 users for data
- 100Gbps bandwidth per 8 users
- Both Linux and Windows Virtual Machines shall have access to the data hosted on the parallel filesystem.

#### INFRASTRUCTURE SERVERS

46. Bidder shall provide at least two (2) login nodes, each with below minimum specifications:

- Dual X86 CPUs, each with minimum of 16 physical CPU cores, and at least 256GB DDR5 RAM
  - **Local boot disk:** 1x480GB (or higher) SAS SSD RI use, 1DWPD
  - **Network:** 25Gbps (or higher)
  - **Operating System:** Red Hat Enterprise Linux (RHEL)
47. Bidder shall provide a data migration capability (software and hardware) to allow the Supercomputer users to copy data from the Remote Computing Facility Central High Performance Storage to UDC Computing Facility High Performance Storage, and vice versa, **at a minimum rate of 25GB/s** using multiple parallel rsync sessions. At minimum Bidder shall provide two (2) data mover nodes, for copying data between this **Remote Computing Facility** and Saudi Aramco UDC or DAMMAM-7 facilities, each node with below minimum specifications:
- Dual X86 CPUs, each with minimum of 48 physical CPU cores, and at least 512GB DDR5 RAM
  - **Local boot disk:** 1x480GB (or higher) SAS SSD RI use, 1DWPD
  - **Network:** 200Gbps (2x100GbE or 1x200GbE)
  - **Operating System:** Red Hat Enterprise Linux (RHEL)
48. Bidder shall provide at least three (3) License Servers to host FlexNet services with the below minimum specifications:
- Dual X86 CPUs, each with minimum of 32 physical CPU cores, and at least 256GB DDR5 RAM
  - **Local boot disk:** 1x480GB (or higher) SAS SSD RI use, 1DWPD
  - **Network:** 25Gbps (or higher)
  - **Operating System:** Red Hat Enterprise Linux (RHEL) or Windows
  - Note: these servers can be virtualized provided proposed solution is supported by FlexNet
49. Bidder shall provide at least two (2) Application Servers to host seismic project management application:
- Dual X86 CPUs, each with minimum of 32 physical CPU cores, and at least 256GB DDR5 RAM
  - **Local boot disk:** 1x480GB (or higher) SAS SSD RI use, 1DWPD
  - **Network:** 25Gbps (or higher)
  - **Operating System:** Red Hat Enterprise Linux (RHEL) or Windows
  - **Local data disk:** usable capacity of 7TB NVMe (or higher) (RAID-5)
50. Bidder shall provide at least ten (10) Seismic Processing Platform Servers:
- Dual X86 CPUs, each with minimum of 32 physical CPU cores, and at least 512GB DDR5 RAM
  - **Local boot disk:** 1x480GB (or higher) SAS SSD RI use, 1DWPD
  - **Local data disk:** usable capacity of 28TB (or higher) (RAID-5) – NVMe
  - **Network:** 100Gbps (or higher)
  - **Operating System:** Red Hat Enterprise Linux (RHEL)

#### OTHER NODES

51. Bidder shall provide the required enough servers to support the environment information security and facilitate user authorization and authentication such as **IPA/DNS, NTP, Log Server(s)**, and any required authentication services as required. The Bidder shall provide the details of such servers in their Technical proposal.

52. Bidder shall provide enough nodes for the management, configuration, and administration, visualization services required by the **provided solution**. These nodes can be physical or virtual, based on the OEM recommendation. If virtual nodes are provided, Bidder is responsible to provide capable hosting physical nodes separate from the above supercomputer or storage nodes.

## STORAGE TECHNICAL SPECIFICATIONS

### Parallel Filesystem

53. **Filesystem:** the proposed parallel filesystem shall be either *Enterprise Luster* (from HPE or DataDirect Networks) or *IBM Spectrum Scale*. Any alternatives shall be approved by Saudi Aramco first.
54. **Capacity:** 100,000TB usable disk storage capacity (minimum)
55. **Performance:**
- **Aggregate (minimum):** 1000GB/s for read, and 1000GB/s for write
  - **Single Stream per node (minimum):** 4GB/s for read, and 4GB/s for write.
  - **Multiple Stream per node (minimum):** 10GB/s for read, and 10GB/s for write.
56. **Data Protection:** The storage for data shall be protected against disk failure by at least two levels of parity. The proposed storage solution shall implement a RAID technology that can protect and safeguard the data against a hard disk failure and a second hard disk media error occurring simultaneously. The Bidder shall provide details on how this can be accomplished.
57. The storage for **metadata** shall be based on enterprise class NVMe drives and protected against disk failure using mirroring/RAID.
58. **Data at rest encryption:** shall be supported.
59. **NFS Services:**
- Bidder shall provide two (2) nodes to serve as NFS services to multiple clients within and outside the system. The configuration of these nodes shall be supported by the filesystem manufacturer and meet or exceed the acceptance tests.
60. **Quota:**
- The proposed filesystem shall support user, group, project and pool quotas
61. **Audit Accounting:**
- The proposed filesystem shall support filesystem audit accounting

### Application storage (NAS)

62. Bidder shall provide additional storage capacity that will be used for shared application data and is less I/O intensive activities. This system shall meet the below requirements (as minimum):
63. **Usable Capacity:** 500TiB
64. **Aggregate performance:** 5GB/s read, 1GB/s write
65. **Data at rest encryption:** shall be supported.
66. **Snapshot:** required
67. **Quota:** user and group quota
68. **Audit Accounting:** shall be supported
69. **Note:** This can be on the parallel filesystem storage (as separate filesystem), if snapshot is supported. Bidder needs to provide details on the proposed solutions

## Network Fabric

70. Bidder shall provide an integrated, dedicated high speed network infrastructure that can deliver the required system performance with a subscription ratio no more than 4:1

71. Bidder shall provide details on the following:
- Network diagram showing the proposed solution network design, including connections points (node to switch, and switch to switch) and fixed bandwidth at each point
  - Total aggregate bandwidth
  - sustained node to node bandwidth,
  - the maximum number of connections per nodes, the throughput per connection, and the network subscription ratio.
72. Bidder shall guarantee constant network bandwidth performance of the full system throughout the contract duration, without any degradation overtime.
73. Bidder shall provide adequate performance monitoring tools that can be used by the support and proponent for monitoring interconnect health and performance.

### **HPC Management and monitoring**

74. Bidder shall provide a full working solution with all required software licenses for the full contract duration, including but not limited to:
- Operating system
  - Virtualization software (if needed)
  - Docker container and needed software packages for integration with HPC Job Scheduler
  - HPC nodes management software
  - HPC storage management software
  - HPC performance monitoring system (Compute, Storage, and Network)
75. Bidder shall provide information regarding the management of nodes in the HPC solution, including any necessary software components to establish a unified high-performance technical computing system. The management software should offer capabilities for monitoring and managing the entire system, encompassing dynamic provisioning, operating system and software revision control, real-time monitoring, and historical accounting. It is expected that the Cluster Management software will provide monitoring functionalities for the hardware, software, and performance of the HPC solution.
76. Bidder shall furnish further details on the specific components that will be monitored.

### **Software Development Stack**

77. Bidder shall supply a complete Programming Environment. The environment shall include (but not limited to) :
- IDEs, (e.g. Develop, Eclipse, JEdit),
  - C/C++/Fortran Compilers for Intel and AMD processors
  - Totalview Debugger (64 processes)
  - Parallized GDB
  - Valgrind for HPC
  - NVIDIA HPC SDK
  - Performance Analysis and optimization Tools

### **HPC Job Scheduler (Slurm, UGE, PBSPro)**

78. The Bidder shall provide a job scheduler that is topology-aware and energy management-aware.

79. Job scheduler should be fully integrated with the programming environment, IO & libraries, and performance tools.
80. The job scheduler programmability, installation, configuration, maintenance, and upgrade is the responsibility of the Bidder.

### Other System Tools

81. Bidder shall provide Remote visualization licenses
82. Bidder shall supply, install and manage FlexNet License Server Management for managing supercomputer applications licenses. If the Bidder can not comply with this requirement, Bidder can highlight this during JobEx or Q&A phase.
83. Bidder shall be responsible on providing, configuring and maintaining needed compute resources for system authentication and authorization in compliance with Saudi Aramco information security requirements.
84. Bidder shall provide all required system tools needed for the operation of the provided HPC system

**--- END OF PHASE-1 ---**

# Seismic Processing Compute Expansion Project

Phase 2

Option for Additional 55 Petaflops



## REMOTE COMPUTING FACILITY REQUIREMENTS

### General Requirements

1. Bidder shall note that this phase is non-committing to Saudi Aramco. This is phase is a contract provision that may or may not be ordered or may change over time in response to business requirements. However, the cost of this provision will be included in the evaluation of the overall proposal.
2. The term **Remote Computing Facility** includes land, data center space, chillers, fencing, securing perimeter, CCTV cameras, power and cooling as well as office spaces to be used by the Bidder related support requirements.
3. In the case Bidder is building a new facility, the facility shall comply with the Uptime Institute's **Tier-1 standards** (at minimum). The Bidder shall supply certification of this compliance.
4. In the case Bidder is building a new facility, Bidder is required to provided documented proof of either ownership or lease agreement of the proposed data center facility. This documentation shall demonstrate the Bidder's legal right to operate and provide services from the specified location. Please include a copy of the property deed, lease agreement, or any other relevant legal document verifying the Bidder's rights to the facility. Failure to provide this proof may result in the rejection of the proposal.
5. Bidder shall provide a **committed timeline** for resources mobilization, system deployment and configuration, from the award time. The Bidder shall identify a designated project manager at least one week after the award time. The project manager shall develop the project plan document with Saudi Aramco as per the committed timeline. The project manager shall provide a weekly progress update to Saudi Aramco.
6. This provision may be exercised by Saudi Aramco a year after deployment of Phase-1, tentative deployment year of this phase is 2027.

### Physical Security

7. Same as required in Phase-1.

### Network Connection to UDC Network

8. Saudi Aramco shall provide the fiber cable between Saudi Aramco UDC data center and the connection point (Meet-me Room) at the Bidder proposed facility. However, the Bidder is fully responsible to/shall provide all required networking equipment at both sites that is needed to provide IP network connectivity between UDC and the provided solution (Compute and Storage).
9. Bidder shall note that the new expansion will utilize existing networking infrastructure deployed as part of Phase-1

10. Bidder shall provide details on the data transfer protocols and encryption methods employed to ensure secure data movement.
11. Bidder shall confirm that provided resources (Compute, network, and storage) are dedicated for Saudi Aramco, and no one else has access to the resources even if not utilized.

## SUPERCOMPUTER TECHNICAL REQUIREMENTS

### General Compute Technical Requirements

12. All hardware supplied, including, but not limited to, CPU and GPU nodes, shall not have any power or system limitations/capping. The supplied hardware shall be capable of being fully utilized to the maximum rated power capacity, to ensure potential future application optimization can be realized.
13. Bidder shall provide an integrated supercomputing system that provides (at minimum) FP64 HPL performance below:
  - Theoretical HPL (High Performance Linpack) performance ( $R_{peak}$ ) of **55 PFlops**,
  - Sustained HPL performance ( $R_{max}$ ) of **35PFlops** of GPU nodes.
14. Bidder acknowledge that providing a solution that does not meet the above performance requirements and the acceptance tests will require providing additional hardware till all requirements are met or waived by Saudi Aramco.
15. The proposed hardware be **certified and approved** by application owners (SLB, and AspenTech) and/or Saudi Aramco.
16. Bidder shall include all required software licenses to meet the supercomputing requirements in the proposal for the full duration of the contract without any additional cost to Saudi Aramco after the start of the lease agreement.
17. All memory channels on the provided **Compute nodes** shall be fully populated, with minimum DIMM size of 32GB DDR5
18. No limit on the ingress and egress network traffic between new expansion and Saudi Aramco facilities. Saudi Aramco shall have no cap on the data volume moved between the two sites.
19. Bidder shall provide a system with at least **170 GPU** nodes, each node with below **minimum** specifications:
  - **GPU**: below options can be supplied
    1. 8XNVIDIA B100 SXM GPUs HBM3, or better
    2. 8XAMD MI300X Infinity Fabric GPUs HBM3, or better

- **CPU:** below options can be supplied
    1. 2x AMD CPUs that can drive the proposed 8 GPUs at full capacity with total DDR5 RAM capacity that is at least 1.5X total HBM capacity per node, or
    2. 2x Intel CPUs with enough physical cores that can drive the proposed 8 GPUs at full capacity with total DDR5 RAM capacity that is at least 1.5X total HBM capacity per node
  - **Local boot disk:** 1x480GB (or higher) SAS SSD RI use, 1DWPD
  - **Local data disk:** usable capacity of 24TB (or higher) (RAID-0 on multiple drives) providing minimum read of 40GB/s per local filesystem. Bidder can also provide a **central storage** solution that is reliable and secure and can provide dedicated 24TB per 8 GPUs and delivers minimum read of 5GB/s per GPU. The central storage shall be based on NVMe drives with ability to handle aggregate performance of simultaneous I/O from all GPU nodes and maintain the per GPU capacity and performance requirement.
  - **Network:** 800Gbps (100Gbps / GPU) (or more if needed to meet acceptance tests)
  - **Operating System:** Red Hat Enterprise Linux (RHEL)
20. Bidder shall provide a data migration capability (software and hardware) to allow the Supercomputer users to copy data from the Remote Computing Facility Central High Performance Storage to UDC Computing Facility High Performance Storage, and vice versa, **at a minimum rate of 25GB/s** using multiple parallel rsync sessions.
21. Bidder shall provide enough nodes for the management, configuration, and administration services required by the **provided solution**. These nodes can be physical or virtual, based on the OEM recommendation. If virtual nodes are provided, Bidder is responsible to/shall provide capable hosting physical nodes separate from the above supercomputer or storage nodes.

## STORAGE TECHNICAL SPECIFICATIONS

### Parallel Filesystem

22. **Filesystem:** the proposed parallel filesystem shall be either *Enterprise Luster* (from HPE or DataDirect Networks) or *IBM Spectrum Scale*. Any alternatives shall be approved by Saudi Aramco first.
23. **Capacity:** 100,000TB usable disk storage capacity (minimum)
24. **Performance:**
- **Aggregate (minimum):** 650GB/s for read, and 650GB/s for write
  - **Single Stream per node (minimum):** 4GB/s for read, and 4GB/s for write.
  - **Multiple Stream per node (minimum):** 10GB/s for read, and 10GB/s for write.

25. **Data Protection:** The storage for data shall be protected against disk failure by at least two levels of parity. The proposed storage solution shall implement a RAID technology that can protect and safeguard the data against a hard disk failure and a second hard disk media error occurring simultaneously. The Bidder shall provide details on how this can be accomplished.
26. The storage for **metadata** shall be based on enterprise class NVMe drives and protected against disk failure using mirroring/RAID.
27. **Data at rest encryption:** shall be supported.
28. **NFS Services:**
- Bidder shall provide two (2) nodes to serve as NFS services to multiple clients within and outside the system. The configuration of these nodes shall be supported by the filesystem manufacturer and meet or exceed the acceptance tests.
29. **Quota:**
- The proposed filesystem shall support user, group, project and pool quotas
30. **Audit Accounting:**
- The proposed filesystem shall support filesystem audit accounting

#### Network Fabric

31. Bidder shall provide an integrated, dedicated high speed network infrastructure that can deliver the required system performance with a subscription ratio no more than 4:1
32. Bidder shall integrate this network fabric with Phase 1 network fabric. Bidder shall
33. Bidder shall provide details on the following:
- Network diagram showing the proposed solution network design, including connections points (node to switch, and switch to switch) and fixed bandwidth at each point
  - Total aggregate bandwidth
  - sustained node to node bandwidth,
  - the maximum number of connections per nodes, the throughput per connection, and the network subscription ratio.
34. Bidder shall guarantee constant network bandwidth performance of the full system throughout the contract duration, without any degradation overtime.
35. Bidder shall provide adequate performance monitoring tools that can be used by the support and proponent for monitoring interconnect health and performance.

## HPC Management and monitoring

36. Bidder shall provide a full working solution with all required software licenses for the full contract duration, including but not limited to:

- Operating system
- Virtualization software (if needed)
- Docker container and needed software packages for integration with HPC Job Scheduler
- HPC nodes management software
- HPC storage management software
- HPC performance monitoring system (Compute, Storage, and Network)

37. Bidder shall provide information regarding the management of nodes in the HPC solution, including any necessary software components to establish a unified high-performance technical computing system. The management software should offer capabilities for monitoring and managing the entire system, encompassing dynamic provisioning, operating system and software revision control, real-time monitoring, and historical accounting. It is expected that the Cluster Management software will provide monitoring functionalities for the hardware, software, and performance of the HPC solution.

38. Bidder can utilize Phase-1 solution to cover Phase-2 components as long as no major impact on delivered services required in Phase-1. Bidder should furnish further details on the specific components that will be monitored.

## Software Development Stack

39. Bidder shall supply a complete Programming Environment. The environment shall include (but not limited to, if not proposed in Phase-1):

- IDEs, (e.g. Develop, Eclipse, JEdit),
- C/C++/Fortran Compilers for Intel and AMD processors
- Debuggers (comparative debugger, parallized GDB, Valgrind for HPC)
- NVIDIA and/or AMD HPC SDK for GPUs
- Performance Analysis and optimization Tools (if different than Phase-1)

HPC Job Scheduler (Slurm, UGE, PBSPro)

40. The Bidder shall integrate new capacity with Phase 1 HPC Job scheduler.

41. The job scheduler programmability, installation, configuration, maintenance, and upgrade is the responsibility of the Bidder.

Other System Tools

42. Bidder shall be responsible on providing, configuring and maintaining needed compute resources for system authentication and authorization in compliance with Saudi Aramco information security requirements.

43. Bidder shall provide all required system tools needed for the operation of the provided HPC system

**--- END OF PHASE-2 ---**



# Upstream Digital Center

Leading Digital Excellence

